

**SIMATS ENGINEERING  
ASSESSMENT TOOL 1**

**COURSE NAME: Fundamentals of Data Science**

**COURSE CODE: DSA04**

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**COURSE OUTCOME COVERED**

***CO1:** Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

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**QUESTIONS: 20**

**Total Marks:100**

Annexure A

**SHORT ANSWER TEST**

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***Code of Conduct:***

*I G Chandravaradhan Reddy-192425037 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: G Chandravaradhan Reddy***

## RUBRICS FOR CALCULATION

| Criteria                          | Weightage | Level 4 – Exemplary (5 Marks)                                | Level 3 – Proficient (4 Marks)         | Level 2 – Developing (2–3 Marks)             | Level 1 – Beginning (0–1 Mark)  |
|-----------------------------------|-----------|--------------------------------------------------------------|----------------------------------------|----------------------------------------------|---------------------------------|
| <b>Conceptual Understanding</b>   | <b>30</b> | Demonstrates clear and complete understanding of the concept | Good understanding with minor gaps     | Basic understanding with some misconceptions | Poor or incorrect understanding |
| <b>Accuracy of Answer</b>         | <b>25</b> | Completely correct answer with no errors                     | Mostly correct with minor errors       | Partially correct with noticeable errors     | Incorrect or irrelevant answer  |
| <b>Application of Knowledge</b>   | <b>20</b> | Correctly applies concepts to solve the question/problem     | Applies concepts with minor mistakes   | Limited or partially correct application     | No or incorrect application     |
| <b>Completeness of Response</b>   | <b>15</b> | Covers all required parts of the question                    | Covers most parts with minor omissions | Covers only some parts                       | Incomplete or missing response  |
| <b>Clarity &amp; Presentation</b> | <b>10</b> | Clear, concise, and well-organized answer                    | Understandable with minor issues       | Somewhat unclear or poorly structured        | Difficult to understand         |

1. A college wants to identify students who may fail in the final exam. How can Data science help?

A Science can analyse students marks attendance, assignments, and previous performance to predict which students may be at risk of failing. The college can then provide early support to those students.

2. An online shopping website wants to recommend products to customers. which use of Data science is applied here?

A Data science is used for product recommendation. It analyzes customers previous purchases, searches, and preferences to recommend suitable products.

3. A company wants to reduce customer loss. How can Data science support this?

A Data science can analyze customer behaviour and identify customers who may leave the company. The company can then provide offers or improved services to retain them.

4. A supermarket wants to know which products sell more during festivals. How can Data science help?

A Data science can analyze past festival sales data to identify products with high demand. This helps the supermarket maintain sufficient stock during festivals.

5. A farmer wants to improve crop yield. Give one use of Data science

A Data science can analyze soil, weather, rainfall, and crop data to help farmers select suitable crops and improve crop yield.

6. A weather app updates temperature every minute which facet of data is involved?

A The facet is velocity. It refers to the speed at which data is generated, updated and processed.

7. Customer comments on social media are collected for analysis. What type of data is this?

A It is unstructured data because social media comments do not usually follow a fixed rows-and-columns format.

8. A hospital dataset contains missing age values and incorrect phone numbers. Which data quality

issue is present?

A This represents a data quality problem, specifically incomplete and inaccurate data. Missing values make the data incomplete, while incorrect phone numbers make it inaccurate.

9. A company stores large files in Hadoop Distributed File System. Which part of the Big Data Ecosystem is used?

A HDFS (Hadoop Distributed File System) is the storage component of the Hadoop Ecosystem. It is used to store large amounts of data across multiple machines.

10. An organization wants to collect live sensor data from IoT devices. Which Big Data Ecosystem component is suitable?

A Apache Kafka is suitable for collecting and handling real-time streaming data from IoT sensors.